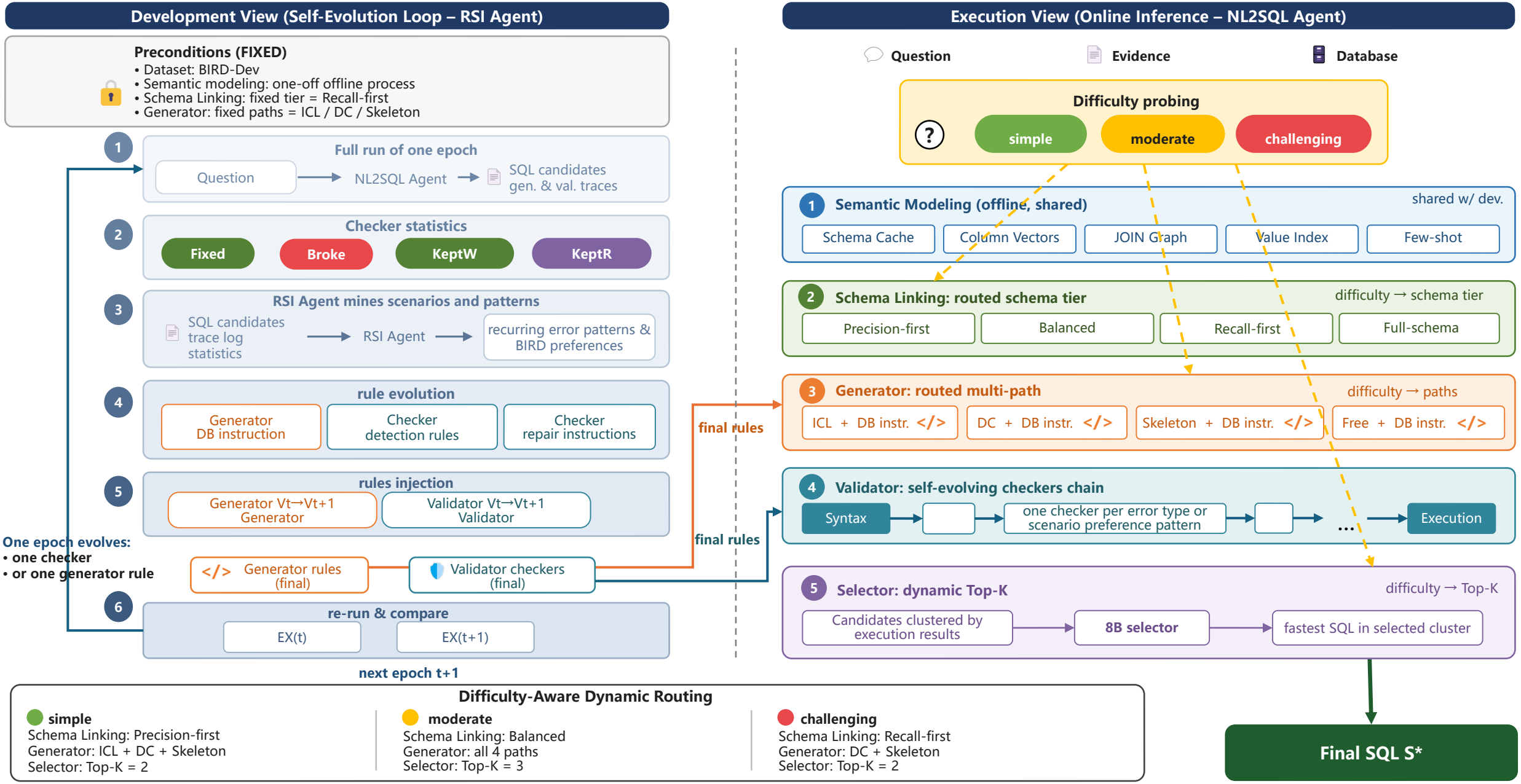

Development View (Self-Evolution Loop – RSI Agent)
Preconditions (FIXED)
• Dataset: BIRD-Dev
• Semantic modeling: one-off offline process
• Schema Linking: fixed tier = Recall-first
• Generator: fixed paths = ICL / DC / Skeleton
1
Full run of one epoch
Question
NL2SQL Agent
SQL candidates gen. & val. traces
2
Checker statistics
Fixed
Broke
KeptW
KeptR
3
RSI Agent mines scenarios and patterns
SQL candidates trace log statistics
RSI Agent
recurring error patterns & BIRD preferences
4
rule evolution
Generator DB instruction
Checker detection rules
Checker repair instructions
5
rules injection
Generator Vt→Vt+1 Generator
Validator Vt→Vt+1 Validator
One epoch evolves:
• one checker
• or one generator rule
</> Generator rules (final)
Validator checkers (final)
6
re-run & compare
EX(t)
EX(t+1)
next epoch t+1
final rules
final rules
Execution View (Online Inference – NL2SQL Agent)
Question
Evidence
Database
Difficulty probing
simple
moderate
challenging
1 Semantic Modeling (offline, shared)
shared w/ dev.
Schema Cache
Column Vectors
JOIN Graph
Value Index
Few-shot
2 Schema Linking: routed schema tier
difficulty → schema tier
Precision-first
Balanced
Recall-first
Full-schema
3 Generator: routed multi-path
difficulty → paths
ICL + DB instr. </>
DC + DB instr. </>
Skeleton + DB instr. </>
Free + DB instr. </>
4 Validator: self-evolving checkers chain
Syntax
one checker per error type or scenario preference pattern
...
Execution
5 Selector: dynamic Top-K
difficulty → Top-K
Candidates clustered by execution results
8B selector
fastest SQL in selected cluster
Final SQL S*
Difficulty-Aware Dynamic Routing
simple
Schema Linking: Precision-first
Generator: ICL + DC + Skeleton
Selector: Top-K = 2
moderate
Schema Linking: Balanced
Generator: all 4 paths
Selector: Top-K = 3
challenging
Schema Linking: Recall-first
Generator: DC + Skeleton
Selector: Top-K = 2